

Calculus II 2024 - Spring.

1.b) $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dz dy dx$

Soln $\int_0^1 \int_0^{1-x} [e^z]_0^{x+y} dy dx = \int_0^1 \int_0^{1-x} (e^{x+y} - 1) dy dx$

$\Rightarrow \int_0^1 [e^{x+y} - y]_0^{1-x} dx \Rightarrow \int_0^1 [e^{x+1-x} - (1-x) - e^x] dx$

$\Rightarrow \int_0^1 (e - 1 + x - e^x) dx \Rightarrow (ex - x + \frac{x^2}{2} - e^x)_0^1$

$\Rightarrow [e - 1 + \frac{1}{2} - e - (0 - 0 + 0 - 1)]$

$\Rightarrow \frac{1}{2} //$

2.a) $y'' - 4xy' + (4x^2 - 2)y = 0 \rightarrow \textcircled{1}$

Let $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \rightarrow \textcircled{2}$

$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$ be $n!^{th}$ of $\textcircled{1}$, $y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

Put y, y', y'' in $\textcircled{1}$,

$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 4 \sum_{n=1}^{\infty} n a_n x^n + 4x^2 \sum_{n=0}^{\infty} a_n x^n - 2 \sum_{n=0}^{\infty} a_n x^n = 0$

Put $k = n-2$ in 1st series, $k = n+2$ in 3rd series

or, $\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - 4 \sum_{n=1}^{\infty} n a_n x^n + 4 \sum_{n=2}^{\infty} a_{n-2} x^n - 2 \sum_{n=0}^{\infty} a_n x^n = 0$

Calculate for $n=0$ and $n=1$

for $n=0$,

$2a_2 x^0 - 2a_0 x^0 = 0$

$a_0 = a_2$

for $n=1$,

$6a_3 x - 4a_1 x - 2a_1 x = 0$

$a_3 = a_1$

for $n \geq 2$,

$(n+1)(n+2) a_{n+2} x^n - 4n a_n x^n + 4a_{n-2} x^n - 2a_n x^n = 0$

or, $(n+1)(n+2) a_{n+2} + 4a_{n-2} = 4n a_n + 2a_n$

or,

$a_{n+2} = \frac{2(a_n(2n+1) - 2a_{n-2})}{(n+1)(n+2)}$

$(n+1)(n+2)$

$$\text{For } n=2, \quad a_4 = \frac{2(a_2(5) - 2a_0)}{12} = \frac{a_0}{2} \quad \left| \quad \text{For } n=3, \quad a_5 = \frac{2(a_3(7) - 2a_1)}{20} = \frac{a_1}{2} \quad \left| \quad \text{For } n=4, \quad a_6 = \frac{2(a_4(9) - 2a_2)}{30} = \frac{a_0}{6} \right.$$

Put coefficients in ②,

$$y = a_0 + a_1x + a_0x^2 + a_1x^3 + \frac{a_0}{2}x^4 + \frac{a_1}{2}x^5 + \frac{a_0}{6}x^6 + \dots$$

$$y = a_0 \left(1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \dots \right) + a_1 \left(x + x^3 + \frac{x^5}{2} + \dots \right) //$$

2.5) i) $x^2 - 4x + 2$ as a Legendre's polynomial by using Legendre formula solⁿ,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n]$$

$$\text{Put } n=0, \quad P_0(x) = \frac{1}{2^0 0!} \frac{d^0}{dx^0} [(x^2 - 1)^0] = 1 \quad \left| \quad \text{Put } n=1, \quad P_1(x) = \frac{1}{2^1 1!} \frac{d}{dx} (x^2 - 1) = \frac{1}{2} \cdot 2x = x \right.$$

Put $n=2$,

$$P_2(x) = \frac{1}{2^2 \cdot 2!} \frac{d^2}{dx^2} [(x^2 - 1)^2] = \frac{1}{8} \frac{d}{dx} [2(x^2 - 1) \cdot 2x] = \frac{1}{2} \frac{d}{dx} [x(x^2 - 1)]$$

$$\Rightarrow \frac{1}{2} (x \cdot 3x + (x^2 - 1)) = \frac{1}{2} (3x^2 - 1)$$

$$\text{Put } n=3, \quad P_3(x) = \frac{1}{2} (5x^3 - 3x)$$

$$\text{Let } f(x) = AP_2(x) + BP_1(x) + CP_0(x)$$

$$x^2 - 4x + 2 = A \frac{1}{2} (3x^2 - 1) + Bx + C$$

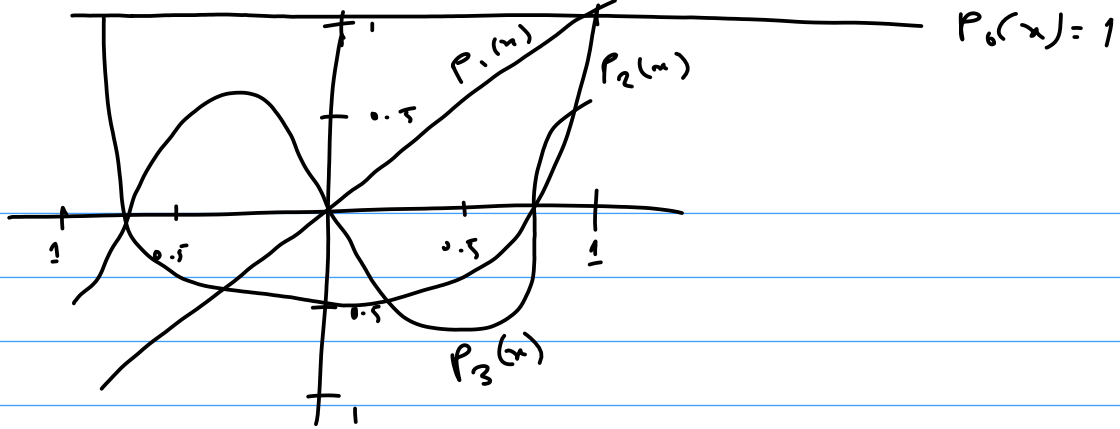
$$x^2 - 4x + 2 = x^2 \left(\frac{3A}{2} \right) - \frac{A}{2} + Bx + C \Rightarrow x^2 \left(\frac{3A}{2} \right) + x(B) + \left(-\frac{A}{2} + C \right)$$

$$\text{Compare, } \frac{3A}{2} = 1 \quad \left| \quad B = -4 \quad \left| \quad C - \frac{A}{2} = 2 \right. \right.$$

$$A = \frac{2}{3} \quad \left| \quad \quad \quad \left| \quad C = \frac{7}{3} \right.$$

$$\therefore x^2 - 4x + 2 = \frac{2}{3} P_2(x) - 4 P_1(x) + \frac{7}{3} P_0(x) //$$

ii)



3. a) $y'' + 2y' + 2y = 5 \sin t$, $y(0) = y'(0) = 0$
Soln)

$$L(y'' + 2y' + 2y) = L(5 \sin t)$$

$$\text{or, } [s^2 L(y) - sy(0) - y'(0)] + 2[sL(y) - y(0)] + 2L(y) = \frac{5}{s^2 + 1}$$

$$\text{or, } L(y)[s^2 + 2s + 2] - s \cdot 0 - 0 - 0 = \frac{5}{s^2 + 1}$$

$$\text{or, } L(y) = \frac{5}{(s^2 + 1)(s^2 + 2s + 2)} \Rightarrow y = L^{-1}\left[\frac{5}{(s^2 + 1)(s^2 + 2s + 2)}\right]$$

$$\text{Sol, } \frac{5}{(s^2 + 1)(s^2 + 2s + 2)} = \frac{As + B}{s^2 + 1} + \frac{(s + D)}{s^2 + 2s + 2}$$

$$\text{or, } 5 = (As + B)(s^2 + 2s + 2) + (Cs + D)(s^2 + 1)$$

$$\text{or, } 5 = As^3 + 2As^2 + 2As + Bs^2 + 2Bs + 2B + Cs^3 + Cs + Ds^2 + D$$

$$\text{or, } 5 = s^3(A + C) + s^2(2A + B + D) + s(2A + 2B + C) + (2B + D)$$

Compare coefficients,

$$A + C = 0 \quad 2A + B + D = 0 \quad , \quad 2A + 2B + C = 0 \quad , \quad 2B + D = 5$$

$$A = -C \quad -4B + B + D = 0 \quad 2B + A = 0 \quad 2B + 3B = 5$$

$$D = 3B$$

$$A = -2B$$

$$\therefore B = 1$$

$$\therefore C = 2$$

$$\therefore D = 3$$

$$\therefore A = -2$$

$$\text{So, } y = L^{-1}\left[\frac{(-2s + 1)}{s^2 + 1} + \frac{(2s + 3)}{s^2 + 2s + 2}\right]$$

$$= L^{-1}\left[\frac{1}{s^2 + 1} - \frac{2s}{s^2 + 1} + \frac{(2s + 3)}{(s + 1)^2 + 1}\right]$$

$$= L^{-1}\left[\frac{1}{s^2 + 1} - \frac{2s}{s^2 + 1} + \frac{2(s + 1) + 1}{(s + 1)^2 + 1}\right]$$

$$= \sin t - 2 \cos t + L^{-1}\left[\frac{2(s + 1)}{(s + 1)^2 + 1} + \frac{1}{(s + 1)^2 + 1}\right]$$

$$= \sin t - 2 \cos t + 2e^{-t} \cos t + e^{-t} \sin t //$$

4.a) $\phi = x^3 + y^3 + z^3 - 3xyz$, $\text{div}(\text{grad } \phi) = ?$

Soln,

$$\text{grad } \phi = \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} = \vec{i}(3x^2 - 3yz) + \vec{j}(3y^2 - 3xz) + \vec{k}(3z^2 - 3xy)$$

$$\nabla \cdot (\text{grad } \phi) = \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} = 6x + 6y + 6z //$$

4.b) $D_a(\phi) = ?$ $\phi = x^2 - y^2 + 2z^2$ in direction of PQ
 $P(1, 2, 1), Q(5, 0, 4)$

$$\text{grad } \phi = 2x\vec{i} - 2y\vec{j} + 4z\vec{k} \text{ at } P(1, 2, 1)$$

$$= 2\vec{i} - 4\vec{j} + 12\vec{k}$$

Direction vector

$$\vec{PQ} = (5-1)\vec{i} + (0-2)\vec{j} + (4-1)\vec{k} = 4\vec{i} - 2\vec{j} + \vec{k}$$

$$\therefore D_a(\phi) = \frac{\nabla \phi \cdot \vec{PQ}}{|\vec{PQ}|} = \frac{(2\vec{i} - 4\vec{j} + 12\vec{k}) \cdot (4\vec{i} - 2\vec{j} + \vec{k})}{\sqrt{4^2 + 2^2 + 1^2}}$$

$$= \frac{(8 + 8 + 12)}{\sqrt{21}}$$

$$= \frac{28}{\sqrt{21}} //$$

4.c) $\vec{F} = (2y+3)\vec{i} + xz\vec{j} + (yz-x)\vec{k}$ find work done $\vec{F} \cdot d\vec{r} = ?$
 $(0, 0, 0)$ to $(2, 1, 1)$

$$x = 2t^2, y = t, z = t^3$$

$$\vec{F} \cdot d\vec{r} = (2y+3)dx + xzdy + (yz-x)dz \quad \left| \begin{array}{l} dx = 4t dt, dy = dt, \\ dz = 3t^2 dt \end{array} \right.$$

$$\vec{F} \cdot d\vec{r} = (2t+3)4t dt + 2t^5 dt + (t^4 - 2t^2)3t^2 dt$$

$$= (8t^2 + 12t + 2t^5 + 3t^6 - 6t^4) dt$$

$$= (3t^6 + 2t^5 - 6t^4 + 8t^2 + 12t) dt$$

Take $y = t$, when $y = 0, t = 0$

$y = 1, t = 1$

$$W = \int_0^1 (3t^6 + 2t^5 - 6t^4 + 8t^2 + 12t) dt$$

$$= \left[\frac{3t^7}{7} + \frac{2t^6}{6} - \frac{6t^5}{5} + \frac{8t^3}{3} + \frac{12t^2}{2} \right]_0^1 \Rightarrow \frac{288}{35} //$$

3.a) Surface integral: ? , $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$, $S: x^2 + y^2 = 1$ in 1st octant.
 $z = 0$ to $z = 2$

Let $\vec{r}$ be position vector,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

Put $x = \sqrt{1-y^2} = \sqrt{1-u^2}$, $y = u$, $z = v$

$$\vec{r} = (\sqrt{1-u^2})\vec{i} + u\vec{j} + v\vec{k}$$

$$\vec{r}_u = \left(\frac{-u}{\sqrt{1-u^2}}\right)\vec{i} + \vec{j} + 0\vec{k} , \vec{r}_v = 0\vec{i} + 0\vec{j} + \vec{k}$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{-u}{\sqrt{1-u^2}} & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \vec{i} + \frac{u}{\sqrt{1-u^2}}\vec{j} + 0$$

$$\vec{F} = v\vec{i} + \sqrt{1-u^2}\vec{j} + u\vec{k} , \vec{F} \cdot \vec{N} = v + u$$

Taking projection on yz plane, $x = 0$,

$$y = 1 , \quad \text{i.e. } 0 \leq u \leq 1 , \quad 0 \leq v \leq 2$$

$$\int_0^2 \int_0^1 (u+v) du dv \Rightarrow \int_0^2 \left[\frac{u^2}{2} + uv \right]_0^1 dv$$

$$\Rightarrow \int_0^2 \left(\frac{1}{2} + v \right) dv = \left(\frac{v}{2} + \frac{v^2}{2} \right)_0^2 = 3 //$$

5.b) Stoke's Theorem: Let $\vec{F}(x,y,z)$ be a continuously differentiable vector field defined on an open surface S bounded by a closed curve C .

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{N} ds$$

$$\vec{F} = (z, x, y) = z\vec{i} + x\vec{j} + y\vec{k} ; \quad C: \text{upper half of sphere } x^2 + y^2 + z^2 = a^2 \text{ in } xy \text{ plane}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \vec{i}(1-0) - \vec{j}(0-1) + \vec{k}(1-0) \Rightarrow \vec{i} + \vec{j} + \vec{k}$$

Let $\vec{r}$ be position vector,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{r}_x = \vec{i} + 0\vec{j} + 0\vec{k} = \vec{i} , \vec{r}_y = 0\vec{i} + \vec{j} + 0\vec{k} = \vec{j}$$

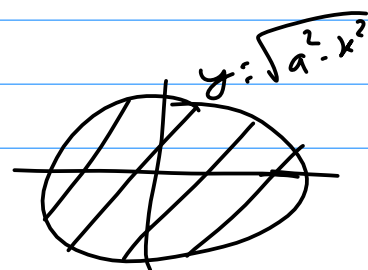
$$\vec{N} = \vec{i} \times \vec{j} = \vec{k}$$

$$(\nabla \times \vec{F}) \cdot \vec{N} = 1$$

Taking projection on xy plane, $z = 0$,

$$y = \pm \sqrt{a^2 - x^2}$$

Put $y = 0, z = 0$; $x = \pm a$



Changing to polar, $0 \leq r \leq a$, $0 \leq \theta \leq 2\pi$

$$\oint \vec{F} \cdot \vec{d}\vec{r} = \int_0^{2\pi} \int_0^a r dr d\theta = \int_0^{2\pi} \frac{a^2}{2} d\theta = \frac{a^2}{2} \times 2\pi = \underline{\underline{\pi a^2}}$$

OR

Gauss divergence theorem:

Let $\vec{F}(x, y, z)$ be a continuous vector fn and having continuous 1st order partial derivative in some domain, then

$$\oint_S \vec{F} \cdot \hat{n} ds = \iiint_V (\nabla \cdot \vec{F}) dV$$

$$\vec{F} = 4x\vec{i} + x^2y\vec{j} - x^2z\vec{k}, \quad s: (0,0,0), (1,0,0), (0,1,0), (0,0,1)$$

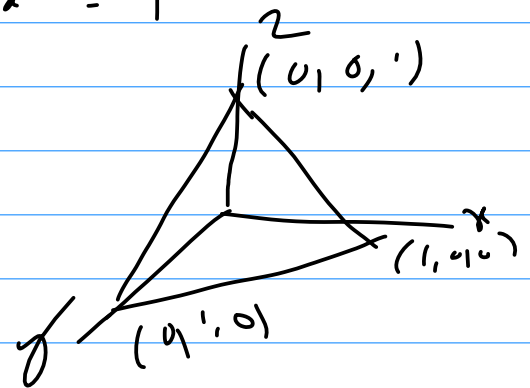
$$(\nabla \cdot \vec{F}) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 4 + x^2 - x^2 = 4$$

$$\text{limit } s: x + y + z \leq 1$$

$$0 \leq x \leq 1 - y - z$$

$$0 \leq y \leq 1 - z$$

$$0 \leq z \leq 1$$



$$\iiint_S (\nabla \cdot \vec{F}) dV = \int_0^1 \int_0^{1-z} \int_0^{1-y-z} 4 dx dy dz \Rightarrow \int_0^1 \int_0^{1-z} 4(1-y-z) dy dz$$

$$\Rightarrow 4 \int_0^1 \left[y - \frac{y^2}{2} - yz \right]_0^{1-z} dz$$

$$\Rightarrow 4 \int_0^1 \left[1-z - \frac{(1-z)^2}{2} - z(1-z) \right] dz$$

$$\Rightarrow 4 \int_0^1 \left[1-z - \frac{(1-2z+z^2)}{2} - z + z^2 \right] dz$$

$$\Rightarrow \frac{4}{2} \int_0^1 (2 - \cancel{2z} - 1 + \cancel{2z} - z^2 - 2z + 2z^2) dz$$

$$\Rightarrow 2 \int_0^1 (1 - 2z + z^2) dz \Rightarrow 2 \left[z - z^2 + \frac{z^3}{3} \right]_0^1$$

$$\Rightarrow \underline{\underline{\frac{2}{3}}}$$

6.a) $f(x) = |x|$; $-2 < x < 2$, $T = 2 - (-2) = 4$, $L = 2$

The Fourier series of $f(x)$ in interval $(-2, 2)$ is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}$$

$$f(x) = \begin{cases} x & ; 0 \leq x \leq 2 \rightarrow \text{even function} \\ -x & ; -2 \leq x \leq 0 \rightarrow \text{odd function} \end{cases}$$

But $f(x)$ is e

$$a_0 = \frac{1}{2L} \int_{-2}^2 f(x) dx = \frac{2}{4} \int_0^2 x dx = \left(\frac{x^2}{2} \right)_0^2$$

$$a_n = \frac{2}{L} \int_0^2 f(x) \cos \frac{n\pi x}{L} dx = \int_0^2 x \cos \frac{n\pi x}{2} dx$$

$$a_n \Rightarrow \left[x \sin \frac{n\pi x}{2} \cdot \frac{2}{n\pi} + \cos \frac{n\pi x}{2} \cdot \frac{4}{n^2 \pi^2} \right]_0^2$$

$$\Rightarrow \left[0 + \frac{4}{n^2 \pi^2} (-1)^n - \frac{4}{n^2 \pi^2} \right]$$

$$\Rightarrow \frac{4}{n^2 \pi^2} \left((-1)^n - 1 \right)$$

$$b_n = \frac{2}{L} \int_0^2 f(x) \sin n\pi x = 0$$

$$\therefore f(x) = 1 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \left((-1)^n - 1 \right) \cos \frac{n\pi x}{2}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{4}{\pi^2} \left[\left(\frac{(-1)^n - 1}{n^2} \right) \cos \frac{n\pi x}{2} \right]$$

For even 'n', $a_n = 0$.

For odd 'n', $a_n = -\frac{8}{n^2 \pi^2}$

$$f(x) = 1 - \frac{8}{\pi^2} \left(\cos \frac{\pi x}{2} + \frac{1}{9} \cos \frac{3\pi x}{2} + \frac{1}{25} \cos \frac{5\pi x}{2} + \dots \right)$$

7. () Show integral is exact,

$$\int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2} (dx - dy + 2z dz)$$

$$\Rightarrow \int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2} dx - e^{x-y+z^2} dy + 2z \cdot e^{x-y+z^2} dz$$

$$F_1 = e^{x-y+z^2}, \quad F_2 = -e^{x-y+z^2}, \quad F_3 = 2z e^{x-y+z^2}$$

$$\frac{\partial F_1}{\partial y} = -e^{x-y+z^2} \quad \left| \quad \frac{\partial F_2}{\partial x} = -e^{x-y+z^2} \quad \left| \quad \frac{\partial F_3}{\partial x} = 2z e^{x-y+z^2} \right. \right.$$

$$\frac{\partial F_1}{\partial z} = 2z e^{x-y+z^2} \quad \left| \quad \frac{\partial F_2}{\partial z} = -2z e^{x-y+z^2} \quad \left| \quad \frac{\partial F_3}{\partial y} = -2z e^{x-y+z^2} \right. \right.$$

$$\therefore \frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} \quad , \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} \quad , \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}$$

$\therefore$ The integral is exact.

Evaluate,

$$\int_{(0,0,0)}^{(2,4,0)} F_1 dx + \int (\text{terms of } F_2 \text{ free from } x) dy + \int (\text{terms of } F_3 \text{ free from } x \text{ and } y) dz$$

$$= \int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2} dx + 0 + 0$$

$$= \left[e^{x-y+z^2} \right]_{(0,0,0)}^{(2,4,0)}$$

$$= \left[e^{2-4} - e^0 \right]$$

$$= e^{-2} - 1$$